ParkMate 🚙

PROTOTYPE ARCHITECTURE REPORT

The Smart Urban Parking-Space Marketplace inspired by Airbnb's high-trust clarity, visual transparency, and seamless interaction design.

Stack: Next.js 16 (App Router) + Supabase (PostgreSQL 15) | **Language:** Strict TypeScript & Zod | **Styling:** Tailwind CSS v4 | **Date:** October 2026

1. Executive Summary & Core Concept

In congested metropolitan areas, urban drivers lose an average of 15 to 20 minutes and significant fuel circling for parking, while private residential driveways, carports, and commercial bays sit vacant during standard business hours.

ParkMate solves this mismatch through a high-trust peer-to-peer and commercial marketplace. It enables homeowners to monetize unused workday parking while giving drivers guaranteed, clearance-verified, and fail-safe reservations near their destinations.

🚙 For Urban Drivers & Commuters

Guaranteed reservations with physical clearance verification, real-time map discovery, digital QR gate passes, 1-click time extensions, and automated booking recovery protection if a reserved spot is blocked.

🏠 For Homeowners & Space Hosts

Automated Smart Space Sharing monetization during idle workday hours (e.g., Mon–Fri 8:30 AM – 5:30 PM), dynamic pricing surge recommendations, draft listing wizards, and complete calendar control.

2. The 5 Core Architectural Differentiators

1. Vehicle Clearance & Compatibility Engine

Unlike generic directories, ParkMate checks vehicle heights, lengths, and widths against garage specs. Low-clearance bays display instant warnings (e.g., *"Exceeds 6.8 ft ceiling limit"*) and safely disable booking to prevent vehicle damage.

2. Guaranteed Booking Recovery Failover

If a driver arrives and finds a space occupied, blocked, or the gate code inoperable, 1-click recovery scans nearby spaces within walking distance, checks vehicle compatibility, and reallocates an equal/upgraded spot at **$0 extra charge** (or issues 100% refund + credit).

3. FlexPark 1-Click Extensions

Drivers running behind schedule can extend reservations (+30m, +1h, +2h) on the fly. The server validates subsequent driver bookings in real-time, executing extensions instantly with zero overlapping conflicts.

4. Smart Space Sharing (Commute Monetization)

Residential hosts monetize driveway downtime during work hours (e.g., Mon–Fri 8:30 AM – 5:30 PM) while keeping the spot reserved for their own car in the evening. The engine analyzes calendar gaps and suggests 1-click publishable slots.

5. Explainable Demand Prediction & Dynamic Pricing AI

A deterministic heuristic model analyzing neighborhood density, rush-hour commuter surges, weekend patterns, and local events to forecast demand scores (0–100) with full transparency, accompanied by an interactive pricing simulator for hosts.

3. Comprehensive List of Features Implemented

Module / Domain	Feature Description	Key Capabilities	Technical Route
Discovery & Search	Split list and map view with real-time vector price pins, vehicle selector, and category filters.	Vector Map Clearance Filter EV / Covered	`/search` `/api/search`
Vehicle Clearance Gate	Hard checks for vehicle dimensions (compact, sedan, SUV, van, truck, EV) against garage heights.	Damage Prevention Custom Dimensions	`compatibility-engine.ts`
Listing Details & Gating	High-res photo carousel, host trust metrics, walkability scores, and server-side calculated price quote.	Privacy Gating Tamper-Proof Price	`/listings/[id]`
Digital Pass & Bookings	Driver wallet showing confirmed reservations, QR access passes, 4-digit gate PINs, and direct map navigation.	QR Code Pass Gate Access PIN	`/bookings` `/api/bookings`
FlexPark Extensions	One-tap reservation extension with automated conflict detection against subsequent bookings.	+1h / +2h Ext Zero Conflict	`/api/bookings/[id]/extend`
Booking Recovery	Arrival failover engine that reallocates nearby compatible spaces if a reserved bay is blocked.	Auto-Reallocation $0 Upgrade	`/api/recovery` `recovery-engine.ts`
Host Hub & Smart Sharing	Owner dashboard tracking listed vs booked hours, space utilization %, and verified vs potential revenue.	Utilization % Slot Recommender	`/owner` `/api/owner/overview`
Dynamic Pricing Simulator	Interactive tool for hosts to test how day, hour, and local events shift demand scores and suggested pricing.	Surge Simulator Event Modifiers	`demand-engine.ts`
Listing Creator Wizard	4-step listing wizard with draft autosave, physical dimension inputs, and photo upload.	Multi-Step Storage Bucket	`/owner/new-listing` `/api/storage/upload`
Saved Spaces	Bookmark favorite spaces with real-time sync across search cards and user profile.	Wishlist Instant Toggle	`/api/saved-spaces`
Idempotent Test Gateway	Safe simulation of credit card authorizations, card declines, and instant refund ledgers.	Idempotent Audit Trail	`payment-engine.ts`

4. Backend Engines & Database Architecture

The core logic of ParkMate is decoupled into specialized domain engines located in `src/lib/`:

booking-engine.ts

State machine transitions (`pending` → `confirmed` → `active` → `completed` / `cancelled` / `recovered`), atomic conflict detection, and tamper-resistant price calculations.

compatibility-engine.ts

Multi-axis dimension matching (height, length, width, side clearance) with safety margins (4 inches vertical, 6 inches horizontal) across 6 vehicle classes.

recovery-engine.ts

Automated failover protocol. Discovers nearest compatible spaces within walking radius, calculates walking times, and waives price differences.

demand-engine.ts

Deterministic surge score engine (0–100) combining base neighborhood density, temporal rush hours (7–9 AM, 4–7 PM), and event flags.

slot-recommendation.ts

Gap analysis scanning owner recurring availability vs bookings to highlight unmonetized peak slots (e.g. Tuesday morning commute).

payment-engine.ts

Idempotent payment transaction orchestrator. Generates payment references, handles test card declines, and manages refund events.

5. Security, Concurrency & Data Integrity

- **PostgreSQL Exclusion Constraints:** `EXCLUDE USING gist (space_id WITH =, tstzrange(start_time, end_time) WITH &&)` strictly prevents double bookings at the database engine level.
- **Row Level Security (RLS):** Supabase policies ensure hosts only edit their own spaces and drivers only access their own reservations.
- **Rate Limiting:** In-memory sliding-window token bucket limiter protecting sensitive reservation and recovery routes.
- **MIME & File Sanitization:** Image uploads checked against a strict whitelist (JPEG, PNG, WebP) with a 5MB limit and randomized storage paths.
- **Address & PIN Privacy:** Exact addresses and gate PIN codes are stripped from search and listing endpoints until a booking is verified.